exclusive SIP have incentives to, and are likely to, enter the new competing consolidator business.[2117] Entry into the competing consolidator business would provide these exchanges new data processing and dissemination, as well as connectivity, revenues. Additionally, being an entrant in the first wave could give a competing consolidator some first mover advantage—even if small—to capture a part of the market that is currently served by the exclusive SIPs. The incentive to have a first mover advantage will only be available to competing consolidators during the initial registration period. In particular, the provision that temporarily precludes registration once the initial registration period closes would provide an incentive to register early, during the initial registration period.[2118] Furthermore, entry costs are going to be lowest during this initial transition period,[2119] making it more attractive to register before this temporary relief expires.

In the unlikely event that only a single competing consolidator enters the market, very few of the Commission's benefit predictions may materialize. The Commission believes that market participants may receive some benefits such as a degree of latency reduction and some

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2117 One of the commenters, an exchange, expressed an interest in becoming a competing consolidator. See MIAX Letter at 1. Additionally, in the past, the same exchange was an active contender to run one of the exclusive SIPs. The announcement made by the law firm conducting the tender offer stated that "The UTP Operating Committee short-listed four firms as the finalists for the RFP bid: CenturyLink, MIAX Technologies, Nasdaq and Thesys Technologies" available at https://www.jandj.com/sites/default/files/library/UTP_SIP_Processor_Announced_2014.pdf (last accessed Sept. 7, 2020)

2118 See supra Section III.H for a discussion on details of the transition period.

2119 The requirements of Regulation SCI will not apply to competing consolidators during an initial phase in period after the effective date of this rulemaking. See supra Section III.F for a discussion of the amendments to Rule 1000 of Regulation SCI.

cost savings from only needing to connect to a single data provider instead of the two exclusive SIPs. However, overall, most of the predicted benefits depend on the new competing consolidator business being a competitive market, and therefore, will not likely materialize if only a single competing consolidator registers.

The Commission believes that a single competing consolidator scenario is also a low probability outcome. The Commission believes that both of the exchanges operating the exclusive SIPs have strong incentives to enter[2120] the new competing consolidator market because under the amendments, the exclusive SIPs will no longer be the exclusive consolidators and disseminators of market data and this will lead to potential revenue losses for the exchanges operating the exclusive SIPs.[2121] The exchanges operating the exclusive SIPs will be incentivized to enter during the initial registration period to start recouping some or all of their potential losses, because competing consolidators that do not enter during the initial wave will not be able to register and operate until the Commission opens up the registration process again.[2122] Additionally, the two exchanges operating the exclusive SIPs have entry costs, profit potentials, and economic interests similar to each other. Thus, neither exchange may leave the new consolidated market data business entirely to the other one and not pursue the chance to recoup some or all of their potential losses from no longer having the exclusive rights to consolidate and disseminate market data. Finally, as mentioned above, there are several other

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2120 Throughout this section, "entry by an exchange operating an exclusive SIP" refers to either the exchange or one of its affiliates becoming a competing consolidator.

2121 See infra Section V.C.2(d) for a discussion of the indirect costs of the decentralized consolidation model.

2122 See supra Section III.H for a discussion on the transition and initial registration period.

market participants who already have expressed an interest in becoming a competing consolidator, expanding the potential pool of initial entrants.[2123]

b. Likelihood of Two Competing Consolidators and Impact on Benefits

The Commission believes that the likelihood that only two market participants enter as competing consolidators is slightly higher than the likelihood of zero or one.[2124] This will most likely happen if two competing consolidators affiliated with the exchanges operating the exclusive SIPs file to be the two initial entrants,[2125] because the disclosure of their identities will potentially deter other potential competing consolidators from registering. The exclusive SIPs have a lot of experience in data consolidation and dissemination, which might deter other potential competing consolidators from entering. Additionally, while several commenters expressed an interest in becoming competing consolidators, they also listed several issues they see as potential risks.[2126] Most of those risks were about the exchanges operating the exclusive SIPs not creating a level playing field for competing consolidators that are not affiliated with them. To the extent that any potential competing consolidator believes that they cannot compete with the exchanges operating the exclusive SIPs, they might not register as additional competing

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2123 See supra note 2109.

2124 One commenter stated that "[o]ne possibility is that only two consolidators will emerge (the current operators of the CTA and UTP plans)." See Angel Letter at 20.

2125 The Commission believes an outcome of two competing consolidators, where one or both are unaffiliated with either of the exchanges operating the exclusive SIPs, is a very low probability one. This is because, as discussed above, the Commission believes that in such a situation both of the exchanges operating the exclusive SIPs will have incentives to enter. Thus it is unlikely that this would be a two competing consolidator scenario.

2126 See, e.g., ACTIV Financial Letter at 1; McKay Letter at 2; NovaSparks Letter at 1; MIAX Letter at 1 for the risks the commenters state about competing consolidators' ability to compete on level playing field.

consolidators, leaving the market with only two competing consolidators where both are affiliated with the exchanges operating the exclusive SIPs.

In the event that the consolidated market data business is served by only two competing consolidators that are both affiliated with the exchanges operating the exclusive SIPs, some of the economic benefits of the competing consolidator model may be limited. In particular, while this result could produce lower gains in delivery efficiency, innovation, and latency differentials and less competitive pressure on data processing and delivery fees, it could bring some degree of competition and corresponding benefits relative to the exclusive SIP model.

If the only competing consolidators to enter are the exchanges operating the exclusive SIPs, the outcome could be lower gains in data delivery efficiency and innovation, and smaller reductions in data consolidation and dissemination latencies. This may be the case primarily because competing consolidators that are affiliated with the exchanges operating the exclusive SIPs would have conflicting profit incentives. For a portion of the market participants, new consolidated market data and proprietary data could be close substitutes. Thus for competing consolidators serving those market participants, the consolidated market data business may cannibalize profits from their parent company's proprietary data business. In that case, these competing consolidators would have to weigh their potential revenue gains from the competing consolidator business against their parent company's potential losses from the proprietary data business. This prospect would reduce these competing consolidators' incentives to compete in this new business line.[2127] Under this scenario, if the market is being served only by two

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[2127] One commenter stated that just having SIAC and Nasdaq UTP as competing consolidators will not create a very competitive market because it "will do little to encourage innovation or price competition as intended by the Proposal." See MIAX Letter at 4.

competing consolidators both affiliated with the exchanges operating the exclusive SIPs, market participants would lack a consolidated market data product vendor without conflicting profit incentives.

Additionally, if there are only two competing consolidators both with conflicting profit incentives, there may not be strong downward competitive pressure on data processing and delivery fees. The Commission's prediction about any downward pressure on data processing and delivery fees depends on the strength of competition among competing consolidators. Having just two competing consolidators—both affiliated with an exchange, with similar economic incentives, and a shared history of serving the whole SIP data market without competing with each other—could soften competition. Specifically, these two competing consolidators might explore opportunities to differentiate in ways that limit competition, such as offering products in different sets of stocks or capturing completely different segments of the market. If market participants would not see these two competing consolidators' products as viable substitutes, they would not be able to switch between them. And this would remove most of the competitive pressure on data processing and delivery fees.

However, even the scenario with only two competing consolidators affiliated with the exchanges operating the exclusive SIPs will bring some degree of competition and corresponding benefits relative to the exclusive SIP model. Unlike the exclusive SIPs today, the exchange-affiliated competing consolidators will operate under a threat of competition from each other and from other potential entrants if an economic opportunity presents itself. In particular, the exchange-affiliated competing consolidators will still have an economic incentive to target each other's customers by introducing new data products serving those customers' needs. In addition, there will likely be some latency benefits from being able to get a consolidated feed from a

single competing consolidator instead of the two exclusive SIPs. Additionally, market participants might see some decline in their consolidation and dissemination costs for equivalent data.[2128] Finally, if an economic opportunity emerges, perhaps because of supra-competitive prices charged by the existing competing consolidators that are affiliated with the exchanges operating the exclusive SIPs, another market participant might register to become a new competing consolidator, and try to capture those customers with product offerings at lower prices.

c. Likelihood of Three or More Competing Consolidators With at Least One Unaffiliated Third Party Registrant and Impact on Benefits

The Commission believes that the most likely scenario for the new data consolidation business is for there to be three or more entrants, where at least one of the newly registered competing consolidators is not affiliated with either one of the exchanges operating an exclusive SIP or an exchange with a proprietary data revenue stream enough to create conflicting profit incentives.[2129] The Commission believes that this scenario will likely lead to vigorous competition and, as a result, will be enough for the predicted benefits to materialize.

The Commission believes that in addition to the exchanges operating a current exclusive SIP, there are several market participants, such as current third party data aggregators or other

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[2128] One commenter agreed. See Angel Letter at 21.

[2129] For a competing consolidator affiliated with an exchange that has a proprietary data revenue stream, there could be conflicting profit incentives as described in Section V.C.2(a)(ii)b. The degree of this conflicting profit incentive will depend on the size of the proprietary data stream relative to the exchange's overall revenues.

intermediary product and/or service providers or exchanges that do not currently operate an exclusive SIP that would have the capability and incentives to enter the newly created competing consolidator business.[2130] Some of the commenters already expressed an interest in doing so.[2131]

Several current market participants, such as third party data aggregators or other intermediary product and/or service providers in the market data space, have the technical capabilities,[2132] customer base,[2133] and incentives a new entrant would need. Some of the potential competing consolidators that might register to enter this new business line are some of

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2130 One commenter said that "[t]he Commission cites only a handful of entities who sought to become data processors in the context of a guaranteed monopoly." See NYSE Letter II at 19. The Commission agrees and notes that for the 2014 UTP SIP tender 11 intent to bid letters were submitted. Similarly, for the CAT SIP tender over 30 intent to bid letters were submitted. See 2014 UTP SIP tender processor selection announcement, available at https://www.jandj.com/sites/default/files/library/UTP_SIP_Processor_Announced_2014.pdf (last accessed Aug. 12, 2020); 2017 CAT SIP tender processor selection announcement, available at https://financialservices.house.gov/uploadedfiles/11.30.2017_mike_beller_testimony.pdf (last accessed Aug. 12, 2020).

2131 See, e.g., McKay Letter at 2; ACTIV Financial Letter at 1; NovaSparks Letter at 1; MIAX Letter at 1 for an expression of their interest in registering as competing consolidators. See also letter from Douglas A. Cifu, Chief Executive Officer, Virtu Financial, to Vanessa A. Countryman, Secretary, Commission, dated Aug. 28, 2020, available at https://www.sec.gov/comments/sr-nyse-2020-05/srnyse202005-7707480-222891.pdf.

2132 See, e.g., McKay Letter at 1 note 1; ACTIV Financial Letter at 1 note 1; NovaSparks Letter at 1; MIAX Letter at 1.

2133 One of the commenters said that while it would like to contemplate being a competing consolidator, any contender would need a large customer base that the commenter believes it does not have. According to the commenter, the big market data aggregators with an existing large customer base are the ones that can achieve this. The Commenter said that "[d]ominated (sic) market data aggregators, like Bloomberg and Refinitiv, would most likely spread their fixed cost to large customer base in quickest time." See Data Boiler Letter I at 84; Data Boiler Letter II at 1.

the most technically sophisticated industry participants. These market participants are currently operating in adjacent markets (e.g., proprietary data aggregation business), making entry into the new consolidated data business easier. Others currently serve as normalizers of the SIP data for retail investors.[2134] They are experienced in market data processing and dissemination, and already serve a portion of the market.

These potential competing consolidators also have the incentives to enter this new competing consolidator business. For an exchange that does not currently operate an exclusive SIP, competing consolidator business would provide an opportunity to get new data consolidation and dissemination as well as connectivity revenues. For a third party data aggregator, it would be a chance to build upon its existing business. For example, if a current third party data aggregator's main revenue source is its normalized SIP data products, then registering as a competing consolidator would be the most direct way for this data aggregator to continue receiving its revenue stream. Even if a current third party data aggregator is mainly focused on the proprietary data aggregation business, becoming a competing consolidator would be a new revenue source and would not create the same conflicting profit incentives described above.

The Commission believes that if three or more competing consolidators enter the market where at least one of them is not affiliated with either one of the exchanges operating the exclusive SIPs or an exchange with sufficient proprietary data revenue to create conflicting profit incentives, this scenario will lead to vigorous competition and will be enough for the predicted

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[2134] See supra note 1777. Companies that normalize market data take in raw data delivered in a variety of protocols and, using feed handlers, normalize into single protocol different from the one used by the original venue. This way a data user can receive one feed using one streaming protocol. See Vela blog, available at https://info.tradevela.com/definitive-guide-to-market-data#normalisation (last accessed Sept. 17, 2020).

benefits to materialize. The conflicting profit incentives described above stem from a competing consolidator's proprietary data customers switching to use consolidated market data products if the data content and speed of the latter become a viable substitute for them. The conflicting profit incentives would stem from the fact that the exchange would have data content revenues to lose as a result of its competing consolidator's customers switching from proprietary data to consolidated market data. But if an exchange does not have significant data content revenues to lose, they would not have such a conflicting profit incentive. A competing consolidator's revenues will mainly come from its data processing, dissemination, and connectivity services, irrespective of the data content it disseminates. Without conflicting profit incentives, such a competing consolidator will focus on expanding its revenue base by aggressively pursuing consolidated market data product clients and capturing an ever larger market share. As part of this pursuit, this competing consolidator would have an incentive to innovate to gain efficiency and speed in data processing and delivery, reduce its costs, and potentially pass on some of these cost savings to its clients to gain market share. Therefore, even if some of the potential competing consolidators have conflicting profit incentives, if at least one other competing consolidator is free from this conflict, competition will intensify.

Overall, the Commission believes that there will initially be at least two competing consolidators and entry into the competing consolidator market space will likely continue until no economic incentives are left for new entry. The Commission believes that the most likely outcome is three or more competing consolidators with at least one competing consolidator that is not affiliated with either one of the exchanges operating the exclusive SIPs or with an exchange with a proprietary data revenue stream that creates conflicting profit incentives. As the number of competitors increase, the level of competition among them will intensify until no

economic incentive is left for new entry. As discussed below,[2135] intensifying competition will benefit market participants.

One commenter stated that the Commission did not consider a possible scenario of "a relatively large number of high-cost consolidators charging high prices for NMS information."[2136] As discussed above,[2137] the Commission believes that a large number of high-cost competing consolidators will not be an equilibrium outcome, because while competing consolidators will have an incentive to differentiate and capture different segments of the market, they can also offer each other's products if they see an economic opportunity to do so. If a large number of high-cost competing consolidators enter this new business line, over time the more efficient of those will capture market share from the less efficient ones.[2138] This is because more efficient competing consolidators will have lower costs and therefore the ability to charge lower prices to market participants and increase their market share. The ability to differentiate will not change this dynamic, because even as competing consolidators differentiate, this real threat of competition will discipline prices and efficiency in the consolidated market data space and will drive out inefficient competing consolidators.

2135 See infra Section V.C.2(c) for a discussion on the benefits of the decentralized consolidation model.

2136 See Nasdaq Letter IV at 23, 24.

2137 See supra Section V.C.2(b) for a discussion of fees as one of the factors to influence the strength of competition in the competing consolidator business.

2138 In a market, more efficient companies have lower production costs and therefore can lower their prices relative to and capture market share from higher cost, thus more inefficient, companies.